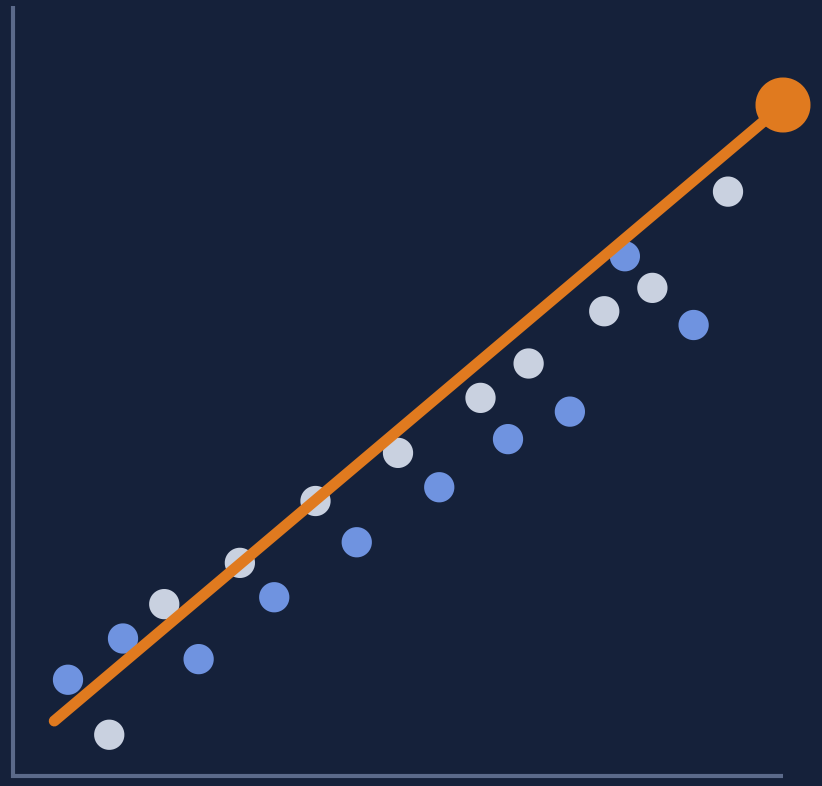
LECTURE 1 · INTRODUCTION TO DATA SCIENCE

Turning data into decisions

Welcome! No coding or statistics background needed, just curiosity.



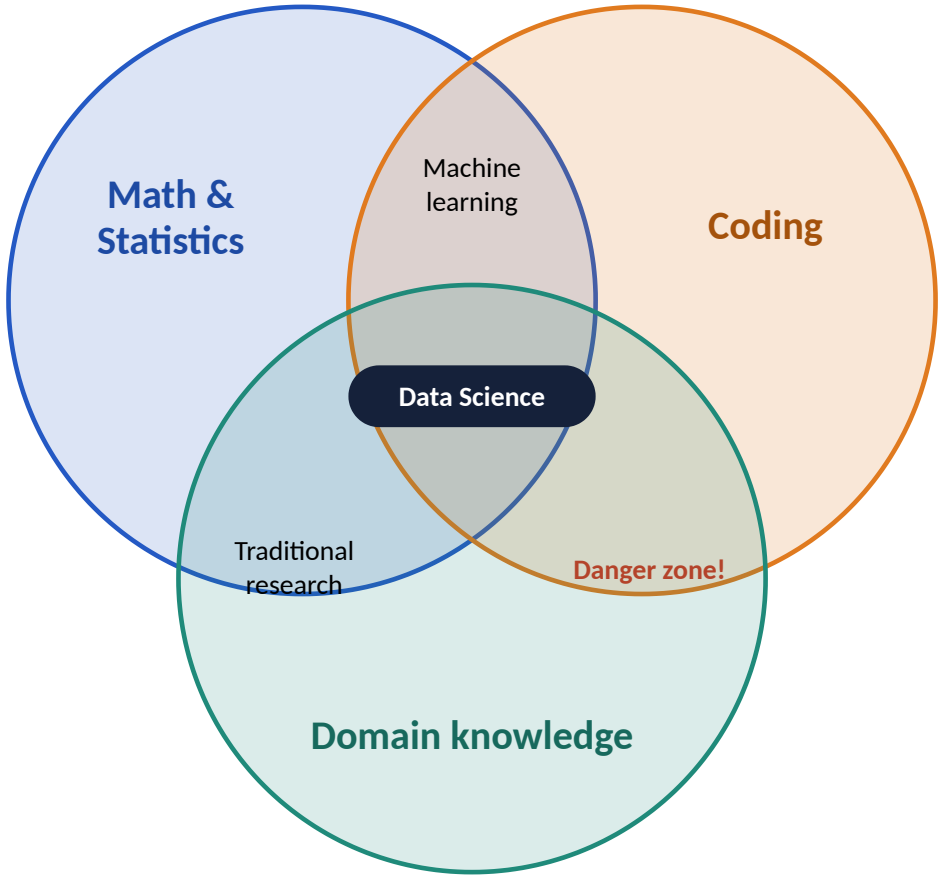
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What is data science?

Using data to answer questions and make better decisions.

- Math & statistics: tell a real pattern from luck
- Coding: handle data too big for a spreadsheet
- Domain knowledge: ask questions that matter

Data science lives where all three overlap.



Adapted from Drew Conway's data science Venn diagram (2010)

What is data?

Most data science starts with a table. Here are four text messages, described as data.

message	length	contains "free"	has a number	label
Claim your FREE gift now	24	yes	no	spam
Class moved to room 305	23	no	yes	ham
Free entry: text WIN to 80086	29	yes	yes	spam
Ok, see you at the library	26	no	no	ham

One row = one example

a single message, patient or sale

One column = one feature

a measured property, used as input (X)

Label column = the answer

what we want to predict (y)

A model learns how the feature columns relate to the label, then predicts the label for new rows. In Lecture 5 this table has 5,572 rows; in Lecture 6 we call the features X and the label y.

Why learn data science?

1

Data is everywhere

Every tap, purchase and scan leaves a data trail.

2

Better decisions

Swap guesses for evidence: what to stock, whom to treat first.

3

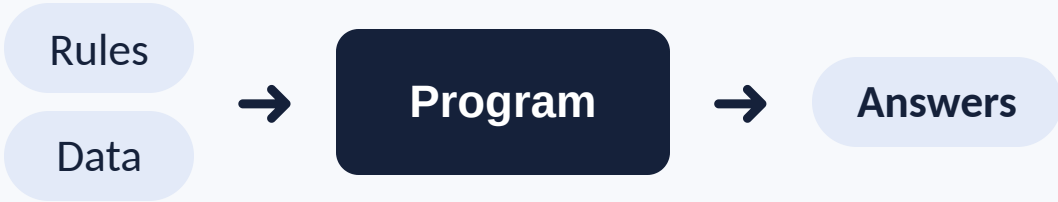
Skills that travel

Health, finance, farming, research: all need people who read data.

Traditional software vs. data science

TRADITIONAL SOFTWARE

A person writes the rules



DATA SCIENCE

The computer learns the rules



	Traditional software	Data science
Example	Calculator, payroll system	Spam filter, face unlock
Output	Exact, the same every time	A prediction, with some uncertainty
Best when	The rules are known exactly	The rules are too complex to write by hand

Live demo: can you write rules that catch spam?

1

Write rules

Keywords like "free" catch 4 of 8 messages. Every fix opens a new hole.

2

Let it learn

A model trained on 30 labeled messages learns its own clues.

3

Try your own

Type any message and see which words look spammy to the model.

The data science lifecycle



A loop, not a line: exploring often sends you back to ask a better question. Modeling is one step of six.

You already use data science every day

1

Recommendations

What will you watch next? Learns from your history and what similar people enjoyed.

2

Healthcare

Does this X-ray look healthy? Learns from past scans labeled by doctors.

3

Finance

Is this payment fraud? Learns from normal spending and past fraud cases.

4

Maps and travel

How long will this trip take? Learns from live speeds of phones on the road.

The road ahead: from zero to your own ML project

Lecture	Topic	You will be able to
1	What is data science?	Explain what data science is and how models learn
2.1 to 2.3	Getting started, Python basics, debugging	Run code offline and online; read and fix errors
3.1 to 3.3	Decisions, lists and tuples, loops	Build a rule-based spam filter, and see it fail
4.1 to 4.2	Functions, dictionaries	Count words and score messages from examples
5	Real data with pandas	Load, clean and explore 5,572 real messages
6	Build a spam detector	Train, test and honestly evaluate a real model
7	Charts and statistics	Summarise and plot data; read a correlation
8 to 10	What is ML? Regression, classification	Fit, predict and compare models
11, 12, Practice 1 and 2	Evaluation, preparing features	Test honestly; use cross-validation and pipelines
13, Practice 3	Clustering, a mini-project	Find groups; run a project from start to end
14, Project	Responsible ML, neural networks, your project	Check fairness; build and present your own model

How this course works

Item	Details
Classes and labs	[Days, times, rooms]. Labs use Jupyter Notebook on the lab PCs (offline)
At home	Google Colab (online), or Jupyter if you install Anaconda on a laptop
Assessment	[Homework __%] · [Quizzes __%] · [Midterm __%] · [Project __%] · [Final __%]
Homework	After every lecture from Lecture 2. Submit your .ipynb file on [LMS / Google Classroom]
Late work	[Late policy]
Help	Office hours: [day, time, room] · Contact: [email]

Using AI tools in this course

1

Allowed

[For example: asking an AI to explain an error message or a concept, like a tutor.]

2

Not allowed

[For example: submitting code or answers generated by an AI as your own work.]

3

Always

You must be able to explain every line you submit. [Say whether and how AI use must be disclosed.]

BEFORE NEXT CLASS

Your first mission

1. Lab: check that Jupyter Notebook opens on your lab PC. Home: open Google Colab if you have internet.
2. Spot one everyday decision that data could improve. Write it in one sentence.
3. Bring one question about data you are curious about.

Office hours: [day, time, room] · Contact: [email]



Questions?

No question is too basic, today especially.

Recap: Lecture 1 in 90 seconds

RECAP

Lecture 1 in 90 seconds

A short recap of today's lecture. Pause any time.

Watch it again at home to revise.